



MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

Department of Electrical & Electronics Engineering

Digital System Design Lab – III Semester

MINI PROJECT REPORT

VEHICLE IMMOBILIZATION SYSTEM

Submitted by

<i>Name</i>	<i>Roll Number</i>	<i>Reg No</i>	<i>Batch</i>
Pisupati Rama Sridhrut	44	230906352	B2
Satwik Abhay	42	230906332	B2
Darshit Chheda	45	230906354	B2

Department of Electrical and Electronics Engineering

Manipal Institute of Technology, Manipal

Aug- Dec 2024

	CONTENTS	Page No.
1.	Abstract /Description about the work	2
2.	Background theory and details about NEXYS 4DDR FPGA	2
3.	Methodology with Block diagram or Flow chart	4
4.	Design/state diagram/truth table	6
5.	Simulation and Implementation Results, Result analysis	8
6.	Contribution of each student	8
7.	References	8

Abstract:

This project implements a vehicle immobilization system using Finite State Machine (FSM) on an FPGA platform. The system ensures vehicle security by only allowing authorized access through a password input. Upon successful password entry, the vehicle is unlocked.

For unauthorized access attempts, an alarm is triggered, and a notification is sent to the vehicle owner. This system is implemented using VHDL, with simulation and verification in Quartus Prime. The design utilizes a state machine to manage various operational states such as idle, password entry, validation, and response.

Background Theory and Details about Nexys4 DDR FPGA

The Nexys4 DDR FPGA board is a versatile digital circuit development platform based on the Xilinx Artix-7 XC7A100T FPGA, optimized for high-performance logic. Here are some key aspects:

1. FPGA Features:

- The Artix-7 XC7A100T FPGA is equipped with 15,850 logic slices, each containing four 6-input look-up tables (LUTs) and eight flip-flops.
- The FPGA provides 4,860 Kbits of block RAM and 240 DSP slices, which support signal processing applications.
- Six clock management tiles with phase-locked loops (PLLs) allow internal clock speeds exceeding 450 MHz.
- An on-chip analog-to-digital converter (XADC) is available for analog signal processing.

2. Memory Configuration:

- The board includes 128 MiB DDR2 SDRAM and a 16 MiB non-volatile SPI flash for storage and configuration.
- The DDR2 memory interface is connected to the FPGA's high-range banks for efficient data transfers

3. Input/Output (I/O) Capabilities:

- The board features 16 user LEDs, 16 switches, and an eight-digit seven-segment display for user interaction.
- Peripherals like an accelerometer, temperature sensor, microphone, and Ethernet PHY support various embedded applications.
- It includes four Pmod connectors for attaching peripheral modules.

4. Programming and Configuration:

- The FPGA can be configured using JTAG, Quad-SPI, USB Host, or microSD card methods.
- Configuration bitstreams can be generated with Xilinx Vivado or ISE tools.
- The Quad-SPI flash allows non-volatile storage for configuration files, enabling automatic reconfiguration on power-up.

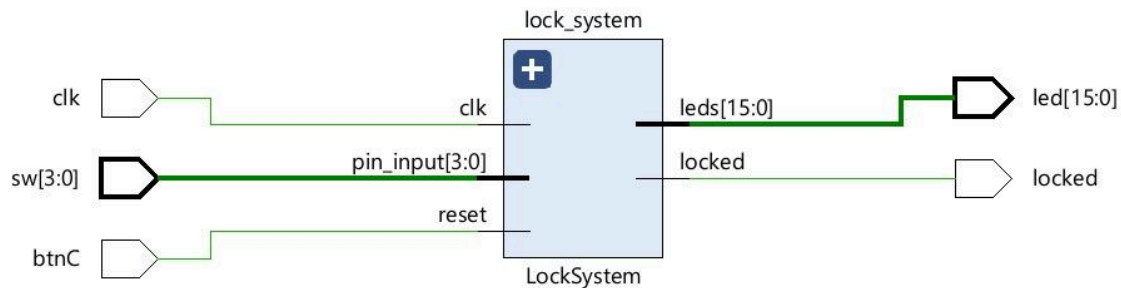
5. Connectivity Options:

- The Nexys4 DDR supports USB-UART for serial communication and JTAG programming.
- A 10/100 Ethernet PHY enables network communication.
- A VGA port allows video output for display applications.

6. Power Supply:

- The board can be powered via USB or an external power supply, supporting voltages from 4.5V to 5.5V.
- Multiple voltage rails (3.3V, 1.8V, 1.0V) are generated on-board to power the FPGA and peripherals.

Methodology with Block diagram :-



METHODOLOGY :-

1. Hardware Requirements:

- Digilent Nexys 4 DDR FPGA Development Board
- USB Cable for programming
- Xilinx Vivado Design Suite

2. Input Interface:

- Utilize 4 switches (SW0-SW3) from the 16 available switches on the Nexys 4 DDR board for PIN input
- The target PIN is 1010 (where 1 represents ON and 0 represents OFF)
- Use a push button (BTNC) as the enter/confirm button

3. Output Interface:

- Use the 16 LEDs (LED0-LED15) available on the board for output display
- When correct PIN: All LEDs illuminate
- When incorrect PIN: Alternate LEDs illuminate (LED0, LED2, LED4, etc.)

4. Design Implementation: a) Module Creation:

- Create a top-level module in VHDL/Verilog
- Define input ports for switches and button

- Define output ports for LEDs

b) State Machine Design:

- Implement a simple state machine with two states:
 - VERIFY_PIN: Check if input matches 1010
 - OUTPUT_RESULT: Control LED pattern based on verification

5. Testing Procedure:

- Simulate design using Vivado simulator
- Test cases:
 - Correct PIN (1010)
 - Various incorrect PIN combinations
 - Button press timing
 - State transitions

6. Implementation Steps: a) Project Setup:

- Create new project in Vivado
- Select Nexys 4 DDR board configuration
- Add design constraints file (.xdc)

b) Pin Assignment:

- Map switches to appropriate FPGA pins
- Map LEDs to corresponding output pins
- Configure button input pin

c) Synthesis and Implementation:

- Run synthesis
- Implement design
- Generate bitstream
- Program FPGA

7. Verification :

- Physical testing on board:
 - Enter correct PIN (1010) → Verify all LEDs light up
 - Enter incorrect PIN → Verify alternate LEDs light up
 - Test button functionality
 - Verify state transitions

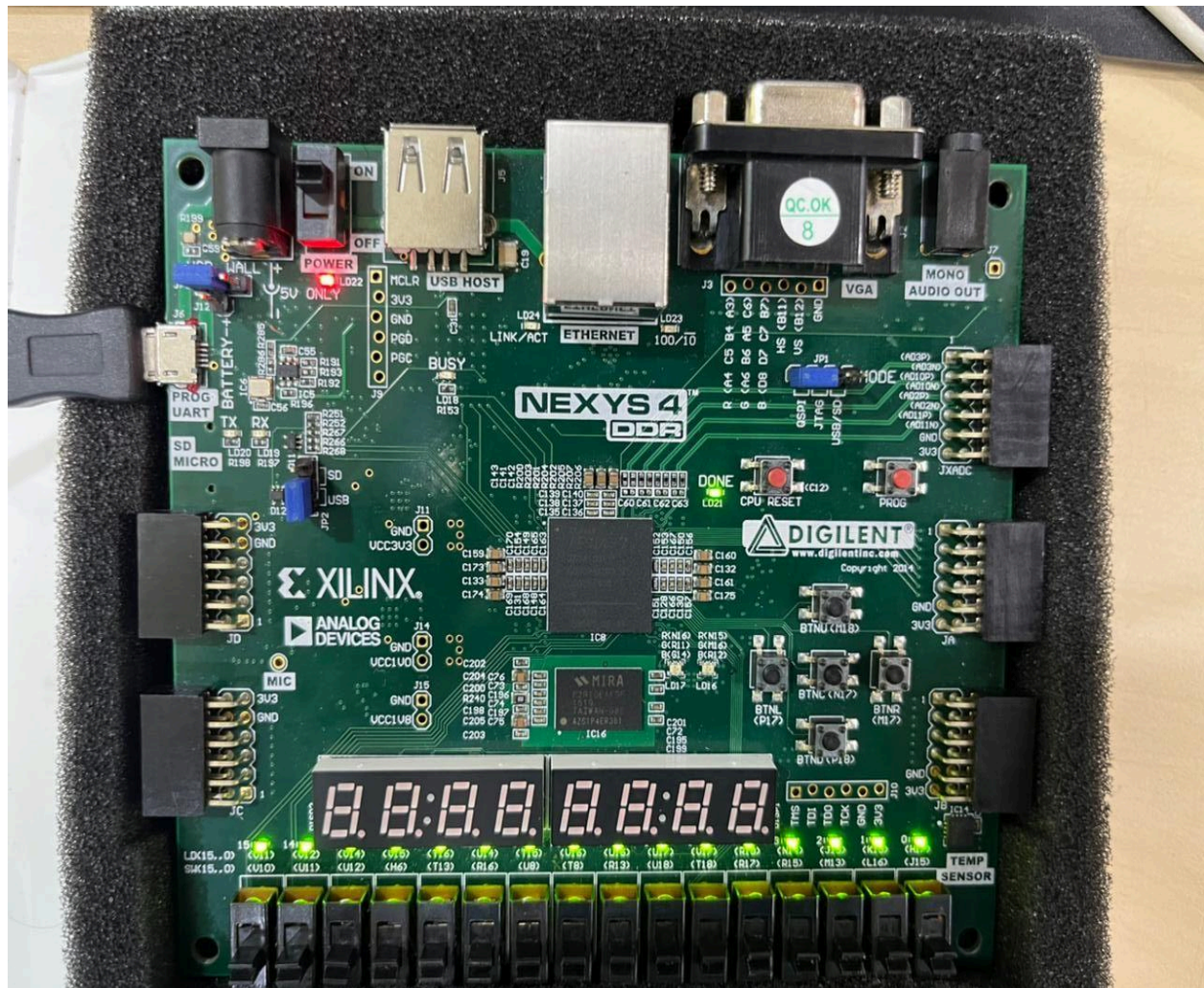
Design/state diagram/truth table :-

reset	confirm_clean	confirm_prev	pin_input	attempt_count	locked (current)	locked (next)	leds (next)	attempt_count (next)
0	X	X	X	X	X	1	16'b1111_1111_1111_1111	4'b0000
1	0	0	X	X	X	locked	leds	attempt_count + 1
1	0	1	X	X	X	locked	leds	attempt_count + 1
1	1	1	X	X	X	locked	leds	attempt_count + 1
1	1	0	predefined_pin	< MAX_ATT EMPTS	1	0	16'b1010_1010_1010_1010	4'b0000
1	1	0	predefined_pin	>= MAX_ATT EMPTS	1	1	16'b0000_0000_0000_0000	attempt_count + 1
1	1	0	!= predefined_pin	< MAX_ATT EMPTS	1	1	16'b1111_1111_1111_1111	attempt_count + 1
1	1	0	!= predefined_pin	>= MAX_ATT EMPTS	1	1	16'b0000_0000_0000_0000	attempt_count + 1
1	1	0	predefined_pin	< MAX_ATT EMPTS	0	0	16'b1010_1010_1010_1010	4'b0000
1	1	0	!= predefined_pin	< MAX_ATT EMPTS	0	1	16'b1111_1111_1111_1111	attempt_count + 1
1	1	0	!= predefined_pin	>= MAX_ATT EMPTS	0	1	16'b0000_0000_0000_0000	attempt_count + 1

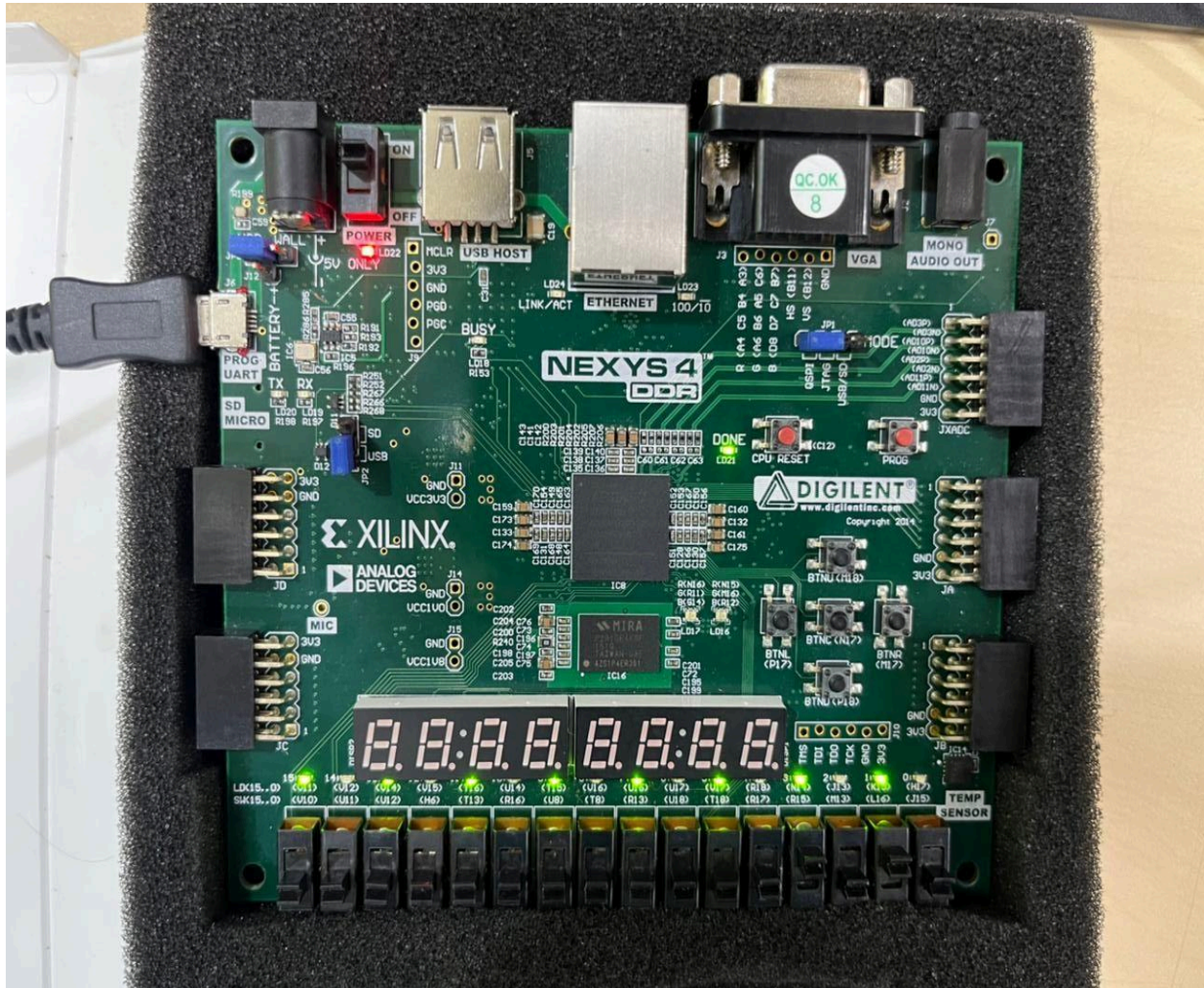
Simulation and Implementation Results, Result analysis:-

Implementation Results :-

1) Locked state:



2) Unlocked State:



Contribution of each student:-

- 1) Pisupati Rama Sridhru:- Contributed to create the Verilog code for the project.
- 2) Satwik Abhay :- Documentation and contributed to system design.
- 3) Darshit Chheda :- Contributed in testing and debugging the program.

References :-

- <https://digilent.com/reference/programmable-logic/nexys-4-ddr/reference-manual>

- <https://www.fpga4student.com/2017/04/simple-debouncing-verilog-code-for.html>